

Theorie 5

5.1.2

$$\underline{p} = \begin{pmatrix} 1 \\ 0.2 \\ 0.5 \end{pmatrix} \quad \hat{d} = \frac{1}{\sqrt{0.3^2 + 0.5^2 + 1}} \begin{pmatrix} 0.3 \\ 0.5 \\ 1 \end{pmatrix} \quad k = \sqrt{0.3^2 + 0.5^2 + 1} \approx 0.1158$$

$$x(t) = \underline{p} + t \cdot \hat{d}$$

$$\text{Grid size: } w = \begin{pmatrix} 0.3 \\ 0.3 \\ 0.3 \end{pmatrix} = h \quad g_{\min} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\text{Indices initial cell: } i_x = \left\lfloor \frac{p_x - g_{\min, x}}{w_x} \right\rfloor = \left\lfloor \frac{1 - 0}{0.3} \right\rfloor = \left\lfloor \frac{10}{3} \right\rfloor = 3$$

$$i_y = \left\lfloor \frac{0.2}{0.3} \right\rfloor = 0$$

$$i_z = \left\lfloor \frac{0.5}{0.3} \right\rfloor = 1$$

$$\text{Start Index: } (3, 0, 1)$$

1. Intersection: because $\hat{d} > 0$ for all 3 components, then follows:

— Distance for next boundary in the corresponding axis:

$$\begin{aligned} \text{for } x: & \Rightarrow g_{\min, x} + (i_x + 1) h_x = p_x + d_x \cdot t_{0,x} \\ & \Leftrightarrow 0 + (3 + 1) 0.3 = 1 + 0.3 \cdot t_{0,x} \\ & \Leftrightarrow 4 - \frac{1}{0.3} = t_{0,x} \\ & \Leftrightarrow \frac{2}{3} = t_{0,x} \end{aligned}$$

$$\begin{aligned} \text{for } y: & \Rightarrow 0 + (0 + 1) \cdot 0.3 = 0.2 + 0.5 t_{0,y} \\ & \Leftrightarrow \frac{1}{5} = t_{0,y} \end{aligned}$$

$$\begin{aligned} \text{for } z: & \Rightarrow 0 + (1 + 1) \cdot 0.3 = 0.5 + 1 \cdot t_{0,z} \\ & \Leftrightarrow \frac{1}{10} = t_{0,z} \end{aligned}$$

Does not need to be normalized

$t_{0,x}$ (also known $t_{\max, x}$)

• Is a scalar value used to determine, when the next cell (in the corresponding Axis) will be crossed.

E.g.: here for z-Axis:

$$\begin{pmatrix} 1 \\ 0.2 \\ 0.5 \end{pmatrix} + \frac{1}{10} \begin{pmatrix} 0.3 \\ 0.5 \\ 1 \end{pmatrix} = \begin{pmatrix} 1.03 \\ 0.26 \\ 0.6 \end{pmatrix}$$

↑
new cell entered

Step Δt :

$$\Rightarrow \Delta t_x = \frac{h_x}{|d_x|}$$

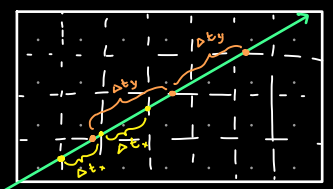
$$\Rightarrow \Delta t_x = \frac{0.3}{0.3} = 1$$

$$\Rightarrow \Delta t_y = \frac{0.3}{0.5} = \frac{3}{5}$$

$$\Rightarrow \Delta t_z = \frac{0.3}{1} = \frac{3}{10}$$

Δt : Scalar value, that measures the distance (if vector, normalized) between two intersections in the corresponding Axis.

• Δt_x remains the same in the grid



Based on $t_{0,x}$ next cell to cross is in the z -Axis (shortest distance)

then with $t_{0,z} + \Delta t_z$ we can get the next intersection

$$t_{1,z} = \frac{1}{10} + \frac{3}{10}$$
$$\Leftrightarrow t_{1,z} = \frac{4}{10}$$

2. Intersection: then, the next closest intersection: $t_{0,x} = \frac{2}{3}$, $t_{0,y} = \frac{1}{5}$, $t_{1,z} = \frac{4}{10}$

y -Axis: $t_{1,y} = \frac{1}{5} + \frac{3}{5}$

$$\Leftrightarrow t_{1,y} = \frac{4}{5}$$

So for the first intersection: (with z -Axis) $t = \frac{1}{10}$

and for the second (y -Axis) $t = \frac{1}{5}$

b) Indices of first four cells

1. Ray starts at: $(3, 0, 1)$

2. Ray intersects z -Axis: $(3, 0, 2)$

3. Ray intersects y -Axis: $(3, 1, 2)$

4. Current $t_{\max, \alpha}$ values: $t_{0,x} = \frac{2}{3}$, $t_{1,y} = \frac{4}{5}$, $t_{1,z} = \frac{4}{10}$ ^{closest}

• So ray will traverse along z -Axis

• Index: $(3, 1, 3)$

• $t_{2,z} = \frac{7}{10}$

5. $t_{\max, \alpha}$ $t_{0,x} = \frac{2}{3}$ ^{closest}, $t_{1,y} = \frac{4}{5}$, $t_{2,z} = \frac{7}{10}$

• Ray intersects x -Axis

• Index: $(4, 1, 3)$

• $t_{1,x} = \frac{5}{2}$

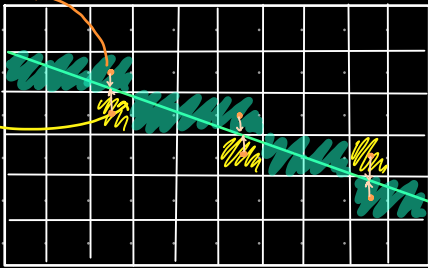
c) Differences with Bresenham's algorithm.

- For the Bresenham's Algorithm, not all cells where the line crosses will be considered. If the slope is roughly 1, then move to $x+1$ and $y+1$. (cells will be skipped)
→ If the line passes through 2 cells, then the closest mid-point of each cell will determine which cell will be drawn.
- Aamides & Woo's Algorithm lists all crossed cells without exceptions.

Bresenham

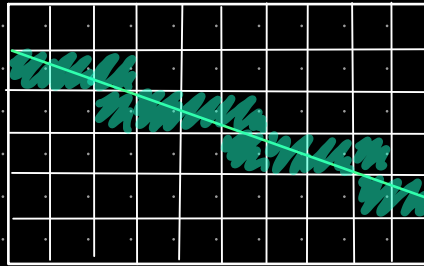
cell's middle point

Skipped cell



Aamides & Woo

No skipped cells



- Bresenham's algorithm is commonly used for line rasterization, whereas Aamides & Woo for Ray traversal, Voxel collision.